BABLU SINGH

To apply strong **Java Back-End** and **Full Stack Engineering** expertise in building and scaling cloud-native, high-performance enterprise systems. Targeting a Senior Software Engineer or Tech Lead–IC role to drive reliability, scalability, and business impact through **Microservices, Distributed Systems**, **DevOps Automation**, event-driven architectures, and AI-assisted development, while contributing as a hands-on technical leader.

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PROFILE SUMMARY

- **Senior Software Engineer** with **over 6 years** of expertise in Java-based back-end and full stack application development, delivering scalable, secure, and high-performance enterprise systems across BFSI, Automotive, and compliance-driven domains.
- Proven specialist in **cloud-native Microservices architecture**, leveraging Spring Boot, Spring Batch, event-driven frameworks, and distributed system patterns to support high-throughput, real-time, and mission-critical workloads.
- Expertise in **managing complete SDLC** including design, development, enhancements, customization, requirement gathering & analysis, testing, coding, customer support and training.
- Strong full stack capability combining **robust Java back-end engineering with modern UI frameworks** (React/Angular), enabling end-to-end ownership of application design, development, and optimization.
- Excellence in **system performance engineering and scalability optimization**, driving significant improvements in throughput, latency, resilience, and availability for large-scale transactional platforms.
- Skilled in **DevOps and CI/CD automation**, orchestrating containerized deployments on Kubernetes, implementing automated pipelines, and enforcing reliability, observability, and security best practices in cloud environments.
- Forward-looking engineer with exposure to **AI-assisted and foundational AI engineering**, integrating LLM-based capabilities and intelligent automation into enterprise applications while mentoring teams on architecture and engineering standards.

CORE COMPETENCIES

- Java Back-End Engineering
- Microservices Architecture & Distributed Systems Design
- Cloud-Native Application Development
- Event-Driven & Real-Time Systems
- Full Stack Development
- API Design, Integration & Development
- System Performance Engineering & Scalability Optimization
- DevOps, CI/CD Automation & Release Engineering
- Enterprise Application Development
- AI-Assisted Engineering & LLM Integration
- Client / Stakeholder Engagement
- Team Building, Training & Management

WORK EXPERIENCE

Since Aug'23: Publicis Sapient, Bengaluru as Senior Software Engineer

Highlights:

- Reduced Spring Batch claim adjudication time from 7 days to 7 minutes by tuning concurrency and optimizing workload distribution.
- Automated Spring Boot version upgrades and implemented continuous vulnerability scanning, raising security compliance by 30%.
- Designed & optimized Java/Spring Boot Microservices processing 14+ million claims with strict real-time performance requirements.
- Deployed HPA-enabled Kubernetes clusters, sustaining 99.99% availability under peak traffic conditions.
- Modernized CI/CD pipelines using GitHub Actions, cutting pipeline execution time by 40%.

Aug'21-Aug'23: Volkswagen IT Services India, Gurugram as Software Engineer

Highlights:

- Earned 3 consecutive Instapat Awards for consistently delivering defect-free, on-time releases.
- Optimized Spring Boot service performance, increasing throughput by 25% and cutting GC pause times by 60 ms.
- Architected Kafka-based streaming pipelines managing 50K+ messages per second with P99 latency below 200 ms to support CO_2 forecasting.
- Enhanced React & Angular front-end performance, improving First Contentful Paint by 20% and reducing Cumulative Layout Shift by 0.1.
- Elevated SonarQube and OWASP compliance scores from 78% to 95% through strengthened code quality and security controls.

- Reduced cloud infrastructure costs by driving Kubernetes migration and application containerization initiatives.

Sept'19-May'21: Capgemini, Mumbai as Senior Analyst

Highlights:

- Developed 5 reusable Java modules adopted across multiple teams, accelerating overall development velocity by 20%.
- Instituted domain-driven design practices, increasing delivery and estimation accuracy by 25%.
- Enforced test-driven development and CI pipelines using JUnit, lowering post-release defect rates by 15%.
- Guided and coached junior engineers on RESTful API design, Microservices architecture, and secure coding standards.

TECHNICAL SKILLS

- **Back-End:** Java, Spring Boot, Microservices, REST APIs, Hibernate, JDBC, Caching, Concurrency, Spring Batch
- **Databases:** Kafka, RabbitMQ, Real-time Processing, Oracle, MySQL, PostgreSQL, MongoDB
- **AI (Foundational Exposure):** MCP Server Development, LangChain, LangSmith, LangGraph, LLM Integration Concepts, Prompting Basics, Agentic AI, Gen AI
- **Front-End:** React, Angular, JavaScript, TypeScript, Responsive UI Development
- **DevOps Tools:** Kubernetes, Docker, Jenkins, CI/CD, GitHub Actions SonarQube, OWASP, Azure, AWS, Monitoring & Observability

PROJECT DETAILS

Title: AI-Powered Multi-Agent Travel Planner (LangGraph + Gemini) **| Period:** 2025 – Present

- Architected an AI-driven travel planning platform leveraging **multi-agent orchestration** to automate trip planning workflow.
- Orchestrated autonomous agents using **Python, LangGraph, and Gemini** to handle flights, hotels, weather intelligence, and dynamic itinerary generation.
- Integrated **real-time aviation APIs (AviationStack)** with intelligent caching and fallback mechanisms to ensure resilient, production-grade recommendations.
- Structured a **modular and extensible agent framework**, enabling scalable enhancements and advanced LLM-based reasoning and decision-making.

Title: AI-Powered JSON Mapping Finder (MCP Server) **| Period:** 2025

- Designed and deployed an **AI-assisted MCP server** to automate JSON schema mapping and data transformation workflow.
- Implemented a **schema-aware parsing engine** capable of identifying deeply nested and complex data structures.
- Devised intelligent mapping strategies using **heuristics combined with embedding-based similarity matching** to maximize mapping accuracy.
- Accelerated ETL and API onboarding processes, **reducing integration effort by up to 90%**.

Title: AI-Powered Wikipedia Search (MCP Server Prototype) **| Period:** 10/2025

- Prototyped an **AI-enhanced search engine** to optimize Wikipedia query interpretation and result relevance.
- Integrated an **MCP server with OpenAI models** to refine semantic understanding and contextual retrieval.
- Elevated search precision and relevance scores, **achieving a 30% improvement in result accuracy**.
- Validated the feasibility of AI-driven search augmentation using cloud-native architecture principles.

Title: CO_2 Emissions Forecast Portal **| Period:** 08/2021 – 08/2023

- Led development of a **real-time CO_2 emissions analytics & compliance platform** aligned with EU regulatory mandates.
- Scaled a **Kafka, Spring Boot & React** streaming architecture to process **50K+ events per second** with **P99 latency under 200 ms**.
- Enabled regulatory forecasting and reporting through high-throughput data pipelines and real-time dashboards.
- Collaborated with a **6-member Agile delivery pod**, contributing across back-end, streaming, and performance optimization layers.

Title: Assembly Line Parts Management System **| Period:** 01/2020 – 05/2021

- Engineered a **web-based logistics and parts management portal** to optimize assembly line operations.
- Streamlined real-time inventory visibility and workflow tracking to minimize production delays.
- Improved overall assembly efficiency by **15%** through enhanced logistics coordination and process transparency.
- Supported operational decision-making with data-driven insights and real-time status monitoring.

EDUCATION

- **B.Tech. in Specialization** from Swami Keshvanand Institute of Technology Management & Gramothan (SKIT) in 2019